

Image Processing Assignment 2

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Task 1 — Problem Statement and Dataset

Two images were selected from the ExDark dataset to contrast in visual content: “Image A” (a room scene with people, furniture, and electronics) and “Image B” (a bottle photographed at night with strong point-light reflections and large uniform dark regions). Both were converted to grayscale before any noise was added, as required. The original colour images are shown below for reference, alongside their grayscale conversions and histograms actually used in Task 2.

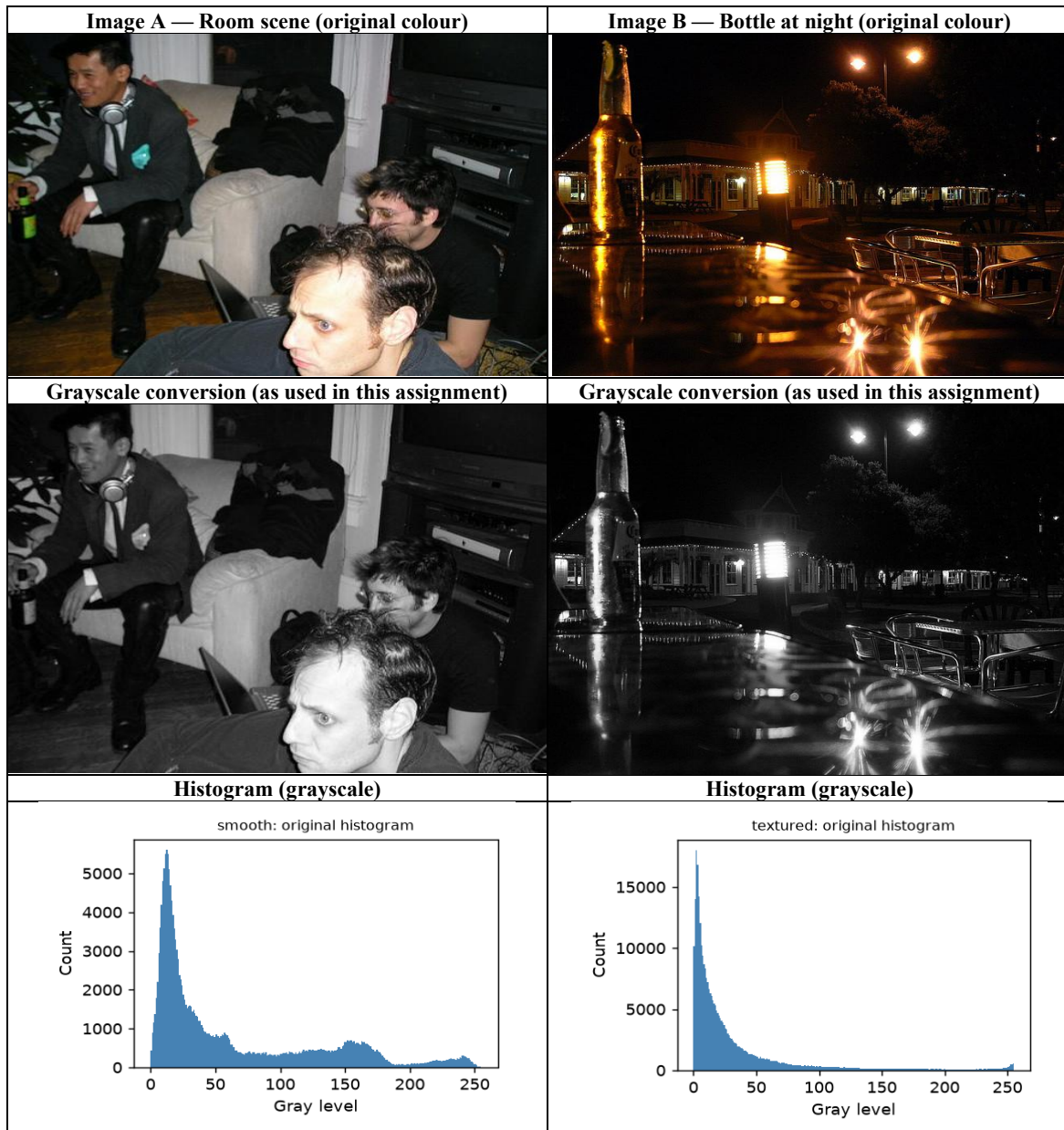


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Task 2 — Noise Models: Images, Histograms, and Noise-vs-PDF Comparison

1. Gaussian Noise

Theoretical PDF: $p(z) = 1/(\sqrt{2\pi}\sigma) \cdot \exp(-(z-\text{mean})^2 / (2\sigma^2))$

Setting: mean = 0, variance = 0.01 (low severity)

Image A — “Room scene” | MSE = 503.10 | PSNR = 21.11 dB

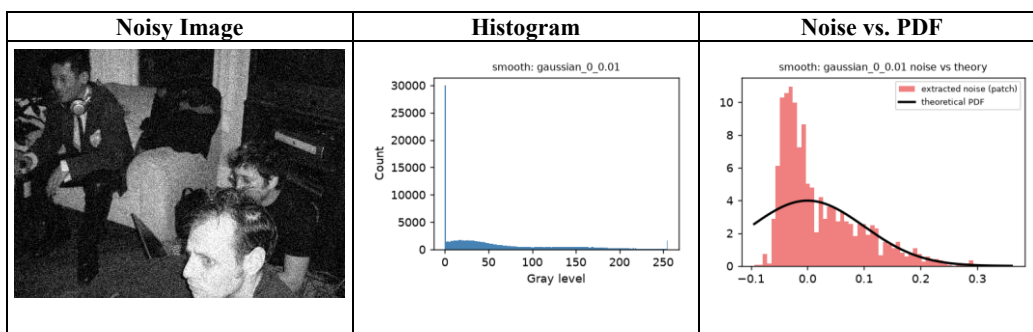
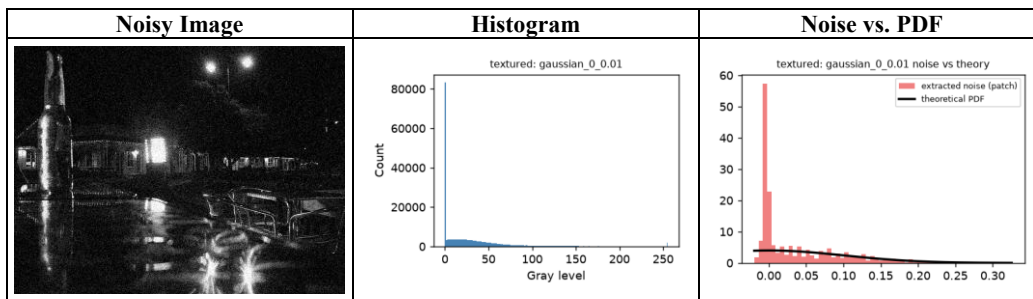


Image B — “Bottle at night” | MSE = 436.69 | PSNR = 21.73 dB



Setting: mean = 0, variance = 0.05 (high severity)

Image A — “Room scene” | MSE = 2161.53 | PSNR = 14.78 dB

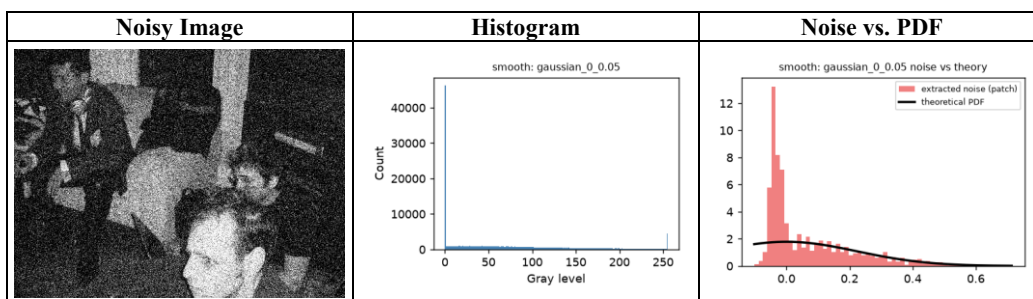


Image B — “Bottle at night” | MSE = 1877.27 | PSNR = 15.40 dB

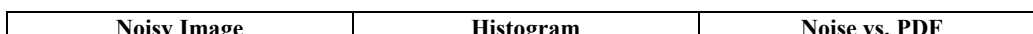
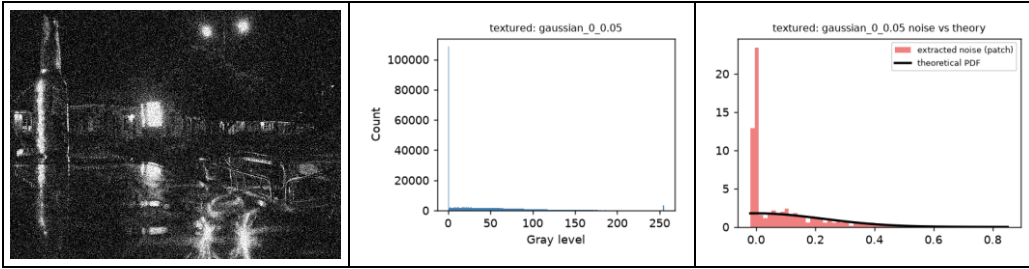


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2. Rayleigh Noise

Theoretical PDF: $p(z) = (z/a^2) \cdot \exp(-z^2/(2a^2))$, $z \geq 0$ (generated via inverse-transform sampling: $z = a \cdot \sqrt{-2 \ln(1-U)}$, $U \sim \text{Uniform}(0,1)$)

Setting: scale $a = 0.1$ (low severity)

Image A — “Room scene” | MSE = 1230.92 | PSNR = 17.23 dB

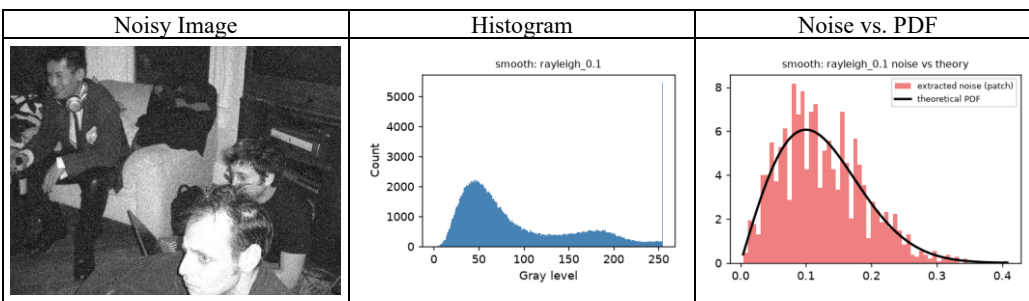
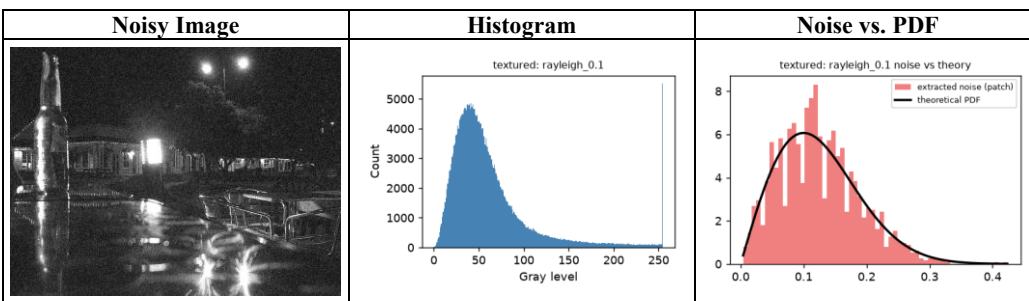


Image B — “Bottle at night” | MSE = 1246.39 | PSNR = 17.17 dB



Setting: scale $a = 0.3$ (high severity)

Image A — “Room scene” | MSE = 9995.19 | PSNR = 8.13 dB

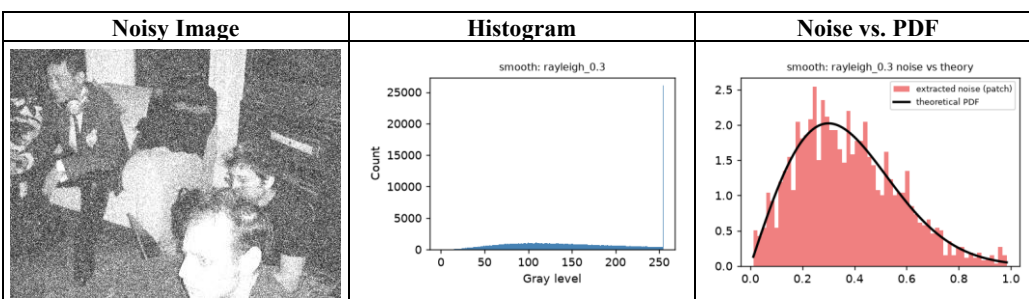


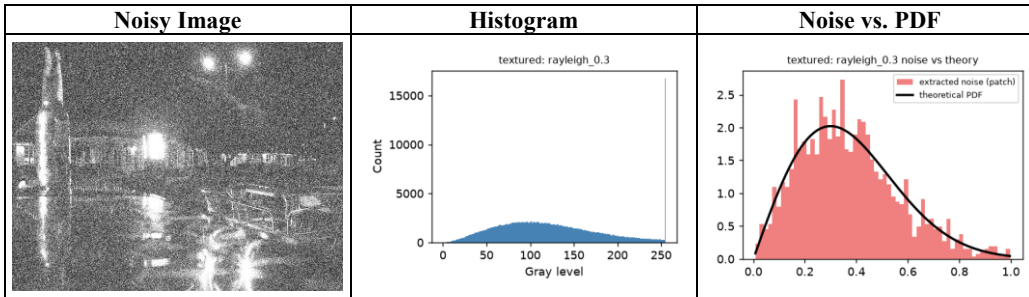
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Image B — “Bottle at night” | MSE = 11008.17 | PSNR = 7.71 dB



3. Erlang (Gamma) Noise

Theoretical PDF: $p(z) = a^b z^{b-1} \exp(-az) / (b-1)!$, $z \geq 0$ (generated via numpy Gamma(shape=b, scale=1/a))

Setting: shape b = 2, rate a = 5

Image A — “Room scene” | MSE = 11297.89 | PSNR = 7.60 dB

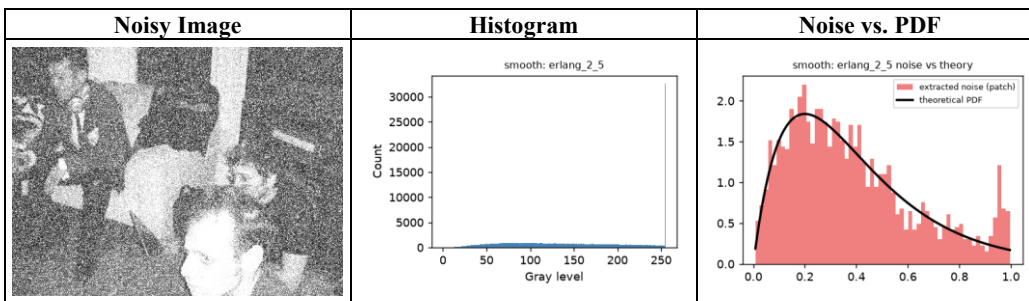
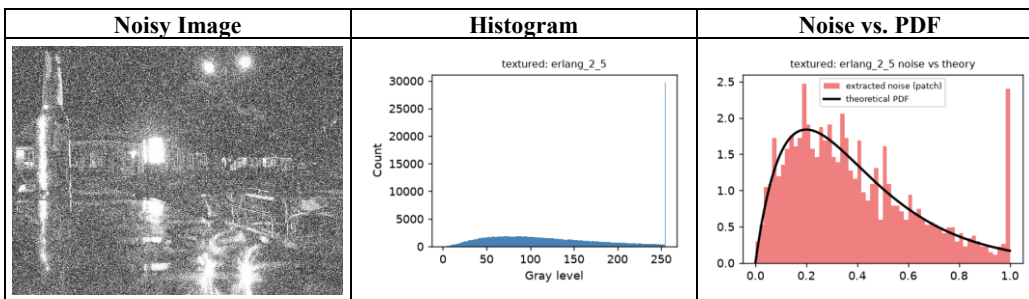


Image B — “Bottle at night” | MSE = 12778.35 | PSNR = 7.07 dB



Setting: shape b = 2, rate a = 10

Image A — “Room scene” | MSE = 3515.43 | PSNR = 12.67 dB

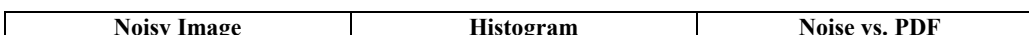


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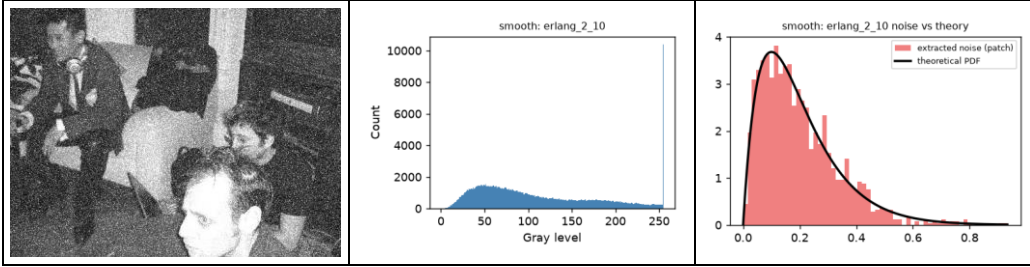
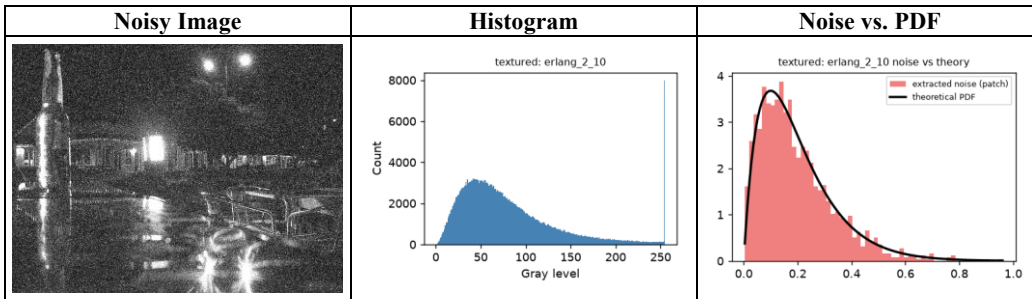


Image B — “Bottle at night” | MSE = 3696.10 | PSNR = 12.45 dB



4. Exponential Noise

Theoretical PDF: $p(z) = a \cdot \exp(-az)$, $z \geq 0$ (generated via numpy `Exponential(scale=1/a)`)

Setting: rate $a = 1$ (high severity)

Image A — “Room scene” | MSE = 23484.12 | PSNR = 4.42 dB

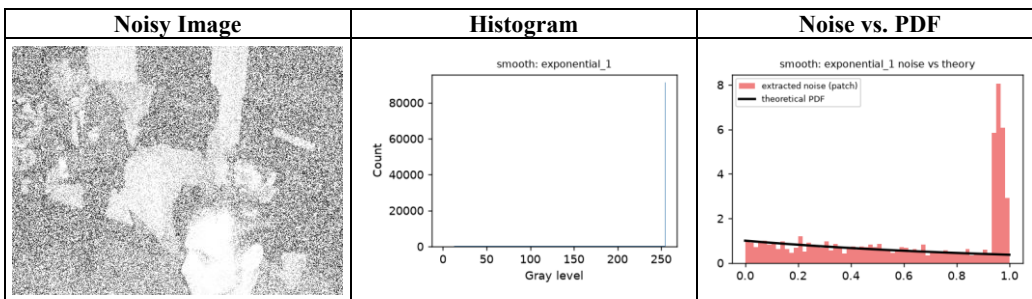
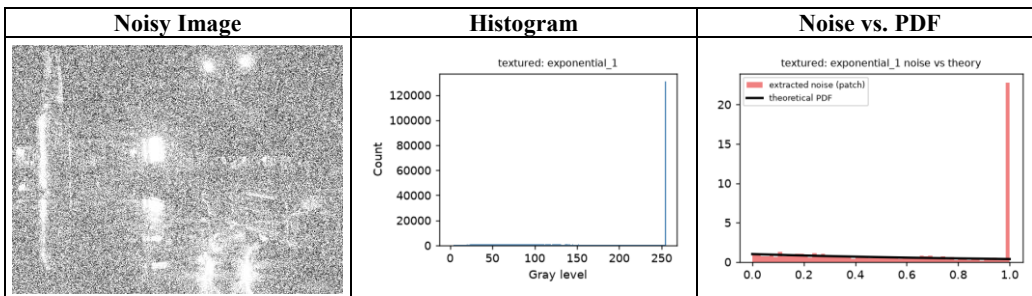


Image B — “Bottle at night” | MSE = 28796.46 | PSNR = 3.54 dB



Setting: rate $a = 3$ (low severity)

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Image A — “Room scene” | MSE = 9109.71 | PSNR = 8.54 dB

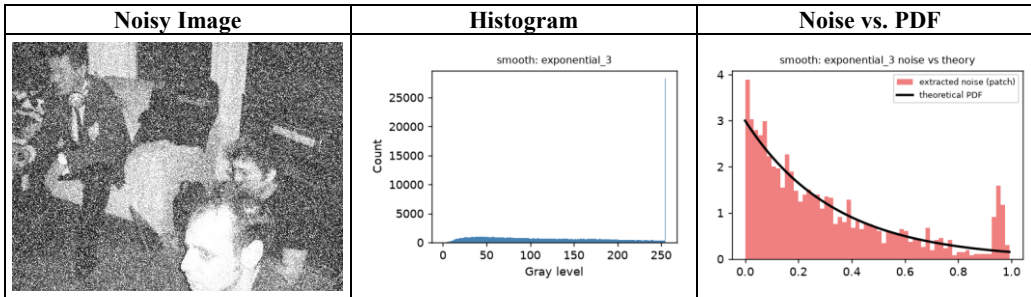
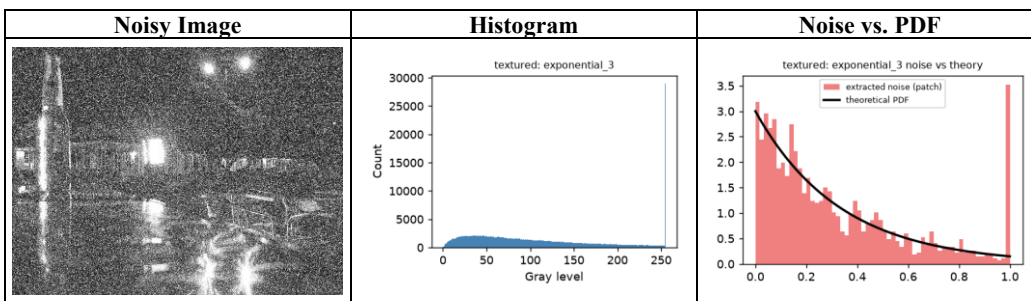


Image B — “Bottle at night” | MSE = 10332.24 | PSNR = 7.99 dB



5. Uniform Noise

Theoretical PDF: $p(z) = 1/(b-a)$ for $a \leq z \leq b$, else 0

Setting: range [-0.1, 0.1] (low severity)

Image A — “Room scene” | MSE = 190.74 | PSNR = 25.33 dB

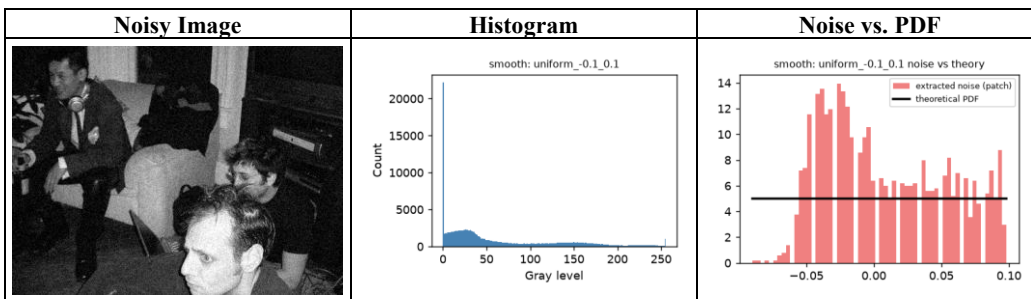


Image B — “Bottle at night” | MSE = 163.74 | PSNR = 25.99 dB

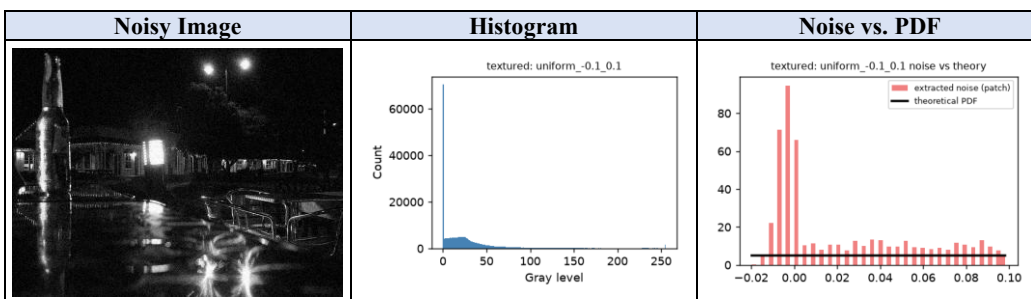


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Setting: range $[-0.3, 0.3]$ (high severity)

Image A — “Room scene” | MSE = 1414.40 | PSNR = 16.63 dB

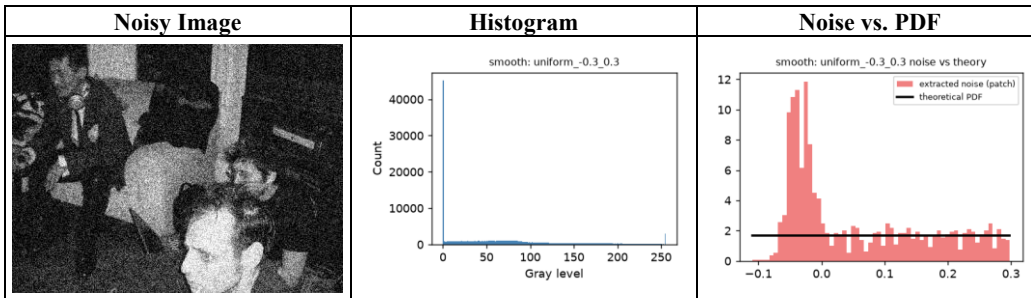
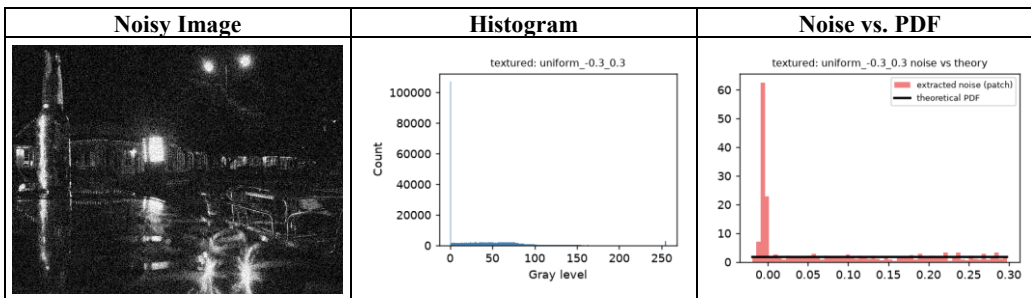


Image B — “Bottle at night” | MSE = 1211.60 | PSNR = 17.30 dB



6. Salt-and-Pepper (Impulse) Noise

Theoretical PDF: $p(z) = P_{\text{pepper}} \cdot \delta(0) + P_{\text{salt}} \cdot \delta(255)$ (impulses only, no PDF curve)

Setting: density $d = 0.05$ (50% salt / 50% pepper)

Image A — “Room scene” | MSE = 1240.73 | PSNR = 17.19 dB

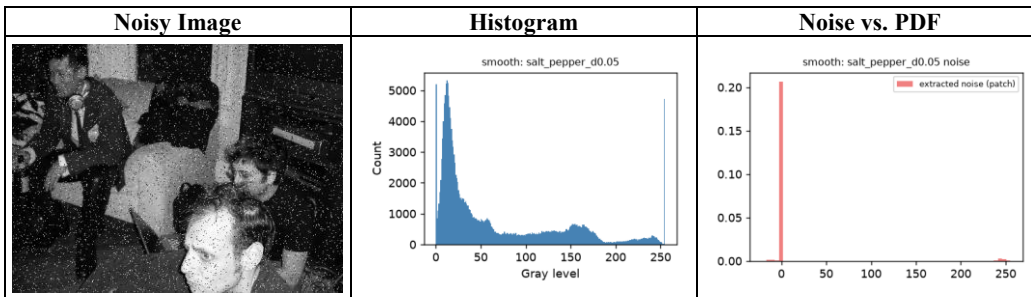


Image B — “Bottle at night” | MSE = 1427.24 | PSNR = 16.59 dB

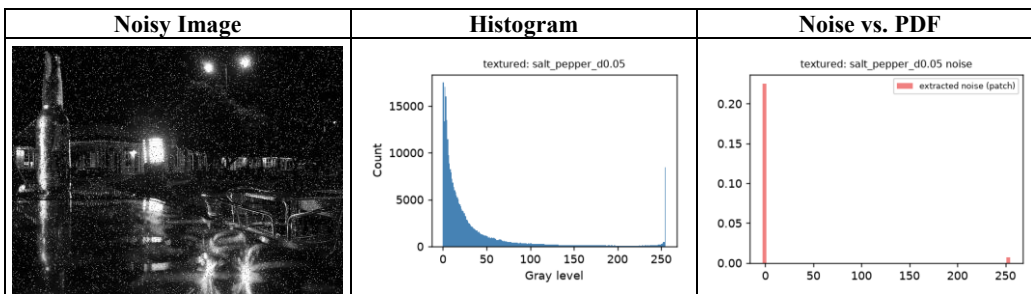


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Setting: density $d = 0.15$ (50% salt / 50% pepper)

Image A — “Room scene” | MSE = 3697.12 | PSNR = 12.45 dB

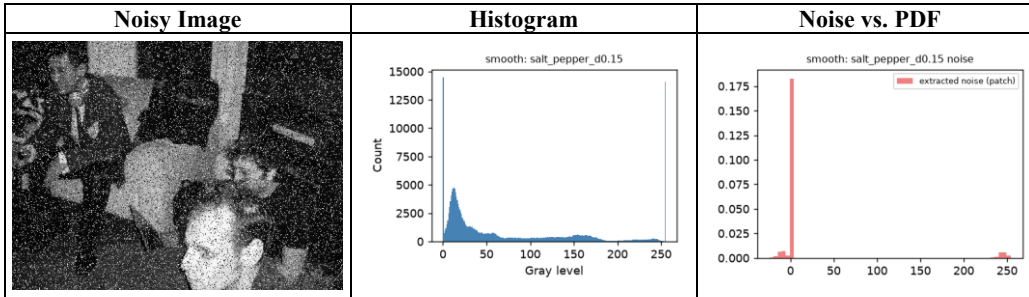
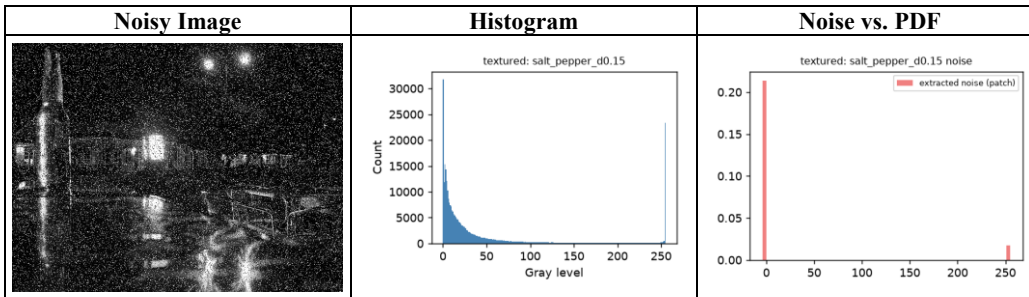


Image B — “Bottle at night” | MSE = 4131.07 | PSNR = 11.97 dB



Summary Table — MSE and PSNR

Image A = room scene (“smooth” pipeline label). Image B = bottle at night (“textured” pipeline label).

Noise Setting	Image A MSE	Image A PSNR (dB)	Image B MSE	Image B PSNR (dB)
mean = 0, variance = 0.01	503.10	21.11	436.69	21.73
mean = 0, variance = 0.05	2161.53	14.78	1877.27	15.40
scale a = 0.1	1230.92	17.23	1246.39	17.17
scale a = 0.3	9995.19	8.13	11008.17	7.71
shape b = 2, rate a = 5	11297.89	7.60	12778.35	7.07
shape b = 2, rate a = 10	3515.43	12.67	3696.10	12.45
rate a = 1	23484.12	4.42	28796.46	3.54
rate a = 3	9109.71	8.54	10332.24	7.99
range [-0.1, 0.1]	190.74	25.33	163.74	25.99
range [-0.3, 0.3]	1414.40	16.63	1211.60	17.30
density d = 0.05 (50% salt / 50% pepper)	1240.73	17.19	1427.24	16.59
density d = 0.15 (50% salt / 50% pepper)	3697.12	12.45	4131.07	11.97

Question 1 — Which noise types are most visually perceptible at similar parameter settings?

Salt-and-pepper noise is the most visually noticeable of all six types even at lower density ($d = 0.05$), since it swaps pixels outright for pure black or white — extreme, high-contrast points the eye picks up instantly, unlike additive noise that only nudges existing values. Among the additive types, Exponential (rate $a = 1$) and Erlang (rate $a = 5$) stand out as most disruptive: both produced the lowest PSNR values above (around 4–8 dB), since their one-sided, heavy-tailed distributions add large, always-positive offsets to many pixels, visibly washing out and brightening the image. Uniform noise at the same nominal range is least perceptible among the

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continuous types, since its bounded, flat distribution never produces the occasional large spike that a Gaussian, Rayleigh, Erlang, or Exponential tail can produce — matching uniform noise's consistently higher PSNR values (25–26 dB at low severity) versus the other continuous types at similar settings.

Question 2 — Why does salt-and-pepper create spikes at extreme gray levels, while the other noise types broaden or shift the histogram?

Salt-and-pepper noise doesn't add a continuous value to each pixel — it replaces a fraction of pixels outright with exactly 0 or 255. Every corrupted pixel lands on one of two values, so its histogram gains two sharp spikes at the extreme gray levels, sitting atop the otherwise unchanged histogram. This shows in the `salt_pepper_d0.05` and `salt_pepper_d0.15` histograms above, where two narrow bars appear at 0 and 255 and grow taller as density rises from 0.05 to 0.15, while the rest of the shape stays largely intact.

Gaussian, Rayleigh, Erlang, Exponential, and Uniform noise, by contrast, all add a continuously distributed value to every pixel's intensity. Since the added value varies smoothly rather than snapping to an extreme, each gray level spreads across a small range of new values, so existing peaks broaden instead of producing new spikes — seen by comparing the `gaussian_0_0.01` and `gaussian_0_0.05` histograms, where the same peak widens as variance increases. Rayleigh, Erlang, and Exponential are also non-negative by definition (PDFs defined only for $z \geq 0$), so the added value is never negative — this shifts the whole histogram brighter in addition to broadening it, most visibly in the `exponential_1` histogram, where the peak pushes noticeably right of the original. Gaussian and Uniform noise, being zero-mean/symmetric here, broaden the histogram without shifting its center.